

Impact of Machine Learning in Business Intelligence (BI) (7 Marks)

Machine Learning (ML) enhances Business Intelligence by enabling systems to automatically analyze large amounts of data, discover hidden patterns, make predictions, and support better business decisions.

Impact of Machine Learning in BI

1. Improved Decision Making

ML analyzes historical and real-time data to provide accurate insights, helping organizations make informed decisions.

2. Predictive Analytics

ML predicts future trends, sales, customer behavior, and market demand using past data.

3. Customer Segmentation

ML groups customers based on purchasing behavior, preferences, and demographics for targeted marketing.

4. Fraud Detection

ML identifies unusual patterns and suspicious transactions, helping businesses prevent fraud.

5. Automation of Data Analysis

ML automatically processes large datasets and generates insights, reducing manual effort.

6. Personalized Recommendations

ML powers recommendation systems that suggest products and services based on customer preferences.

7. Real-Time Business Insights

ML continuously analyzes incoming data and provides real-time reports and dashboards.

8. Risk Management

ML helps identify potential business risks and supports proactive decision-making.

Regression Problems (7 Marks)

Regression is a supervised machine learning technique used to predict continuous numerical values by analyzing relationships between variables. Output variable is continuous. Predicts future values using historical data.

Establishes relationship between dependent and independent variables. Widely used in forecasting and trend analysis.

Types of Regression Problems

1. Simple Regression

Uses one independent variable.

2. Multiple Regression

Uses multiple independent variables.

Applications

- Sales prediction
- House price prediction
- Revenue forecasting
- Stock market prediction
- Demand forecasting

Advantages

- Easy to understand.
- Useful for prediction.
- Helps identify trends.

Limitations

- Sensitive to outliers.
- Assumes relationship exists between variables.

Example

Predicting house prices using area and location.

3. Evaluation of Regression Models (7 Marks)

Evaluation of regression models measures how accurately a model predicts continuous values.

Need for Evaluation

- Determines model performance.
- Compares different models.
- Helps improve prediction accuracy.

Evaluation Metrics

1. Mean Absolute Error (MAE)

Average absolute difference between actual and predicted values.

Lower MAE = Better Model

2. Mean Squared Error (MSE)

Average of squared prediction errors.

Lower MSE = Better Model

3. Root Mean Squared Error (RMSE)

Square root of MSE.

Measures prediction error in original units.

4. R-Squared (R^2)

Measures how much variation in data is explained by the model.

Higher R^2 = Better Model

Advantages

- Measures prediction quality.
- Helps select best model.

4. Linear Regression (7 Marks)

Linear Regression is a supervised machine learning algorithm used to model the relationship between variables using a straight-line equation.

Mathematical Model: $y = mx + c$

Where:

- y = Predicted value
- x = Independent variable
- m = Slope
- c = Intercept

Working

Step 1

Collect historical data.

Step 2

Find the best-fit line through data points.

Step 3

Calculate slope and intercept.

Step 4

Use the equation to predict future values.

Advantages

- Simple and easy to implement.
- Fast training.
- Easy interpretation.

Limitations

- Assumes linear relationship.
- Sensitive to outliers.
- Poor performance on nonlinear data.

Classification Problems (8 Marks)

Classification is a supervised machine learning technique used to assign data into predefined categories or classes.

The output variable is categorical rather than continuous.

The model learns from labeled training data.

After training, it predicts the class of new observations.

Widely used in Business Intelligence for decision support and automation.

Types of Classification

1. Binary Classification

- Involves two possible classes.
- Examples: Spam/Not Spam, Fraud/Not Fraud.

2. Multi-Class Classification

- Involves more than two classes.
- Examples: Product categories, Disease types.

3. Multi-Label Classification

- One observation can belong to multiple classes.
- Example: A news article tagged as both Sports and Business.

Working of Classification

1. Collect labeled data.
2. Train a classification model.
3. Learn patterns and relationships from data.
4. Predict the class of new observations.

Applications

- Email spam filtering
- Customer churn prediction
- Fraud detection
- Medical diagnosis
- Sentiment analysis

Advantages

- Easy to understand and interpret.
- Useful for decision-making.
- Automates categorization tasks.

Limitations

- Requires labeled data.
- Performance depends on data quality.
- Can struggle with imbalanced datasets.

Evaluation of Classification Models

Evaluation of Classification Models is the process of measuring how accurately and effectively a classification model predicts class labels for unseen data. It helps determine whether the model can correctly classify

observations into predefined categories and assess its overall performance.

Need for Evaluation

- Determines model performance.
- Compares different classification models.
- Identifies strengths and weaknesses.
- Helps improve prediction accuracy.

Confusion Matrix

A confusion matrix summarizes prediction results.

Actual / Predicted Positive Negative

Positive	TP	FN
Negative	FP	TN

Where:

- **TP (True Positive):** Correctly predicted positive cases.
- **TN (True Negative):** Correctly predicted negative cases.
- **FP (False Positive):** Incorrectly predicted positive cases.
- **FN (False Negative):** Incorrectly predicted negative cases.

Evaluation Metrics

1. Accuracy

- Measures overall correctness of predictions.
- Higher accuracy indicates better performance.

2. Precision

Measures how many predicted positives are actually positive.

Important when false positives are costly.

3. Recall (Sensitivity)

Measures how many actual positives are correctly identified.

Important when missing positive cases is costly.

4. F1-Score

- Harmonic mean of Precision and Recall.
- Provides a balanced evaluation.

5. Specificity

Measures how well the model identifies negative cases.

Advantages

- Provides detailed performance analysis.
- Helps compare different models.
- Identifies false positives and false negatives.

Limitations

- Accuracy alone can be misleading for imbalanced datasets.
- Multiple metrics are often required for proper evaluation.

1. Bayesian Methods (9 Marks)

Bayesian Methods are machine learning and statistical techniques based on **Bayes' Theorem**, which calculates the probability of an event using prior knowledge and observed data.

$$\text{Bayes' Theorem: } P(A | B) = \frac{P(B|A)P(A)}{P(B)}$$

Where:

- **P(A|B)** = Posterior Probability
- **P(B|A)** = Likelihood
- **P(A)** = Prior Probability
- **P(B)** = Evidence

Based on probability theory.

Updates predictions when new data becomes available.

Uses prior knowledge to improve decision-making.

Commonly used for classification tasks.

Handles uncertainty effectively.

Types of Bayesian Methods

1. Naïve Bayes

Assumes features are independent of each other.

2. Bayesian Networks

Represents probabilistic relationships among variables.

Working of Bayesian Methods (Short)

Step 1: Collect Training Data

Collect labeled data (e.g., Spam and Not Spam emails).

Step 2: Calculate Prior Probability

Find the probability of each class from the training data.

Step 3: Calculate Likelihood

Compute the probability of observing a feature given a class.

Step 4: Apply Bayes' Theorem

Combine prior probability and likelihood to calculate posterior probability.

Step 5: Compare Probabilities

Compare posterior probabilities of all classes.

Step 6: Assign Class Label

Choose the class with the highest probability.

Advantages

- Simple and fast.
- Works well with small datasets.
- Handles noisy data effectively.
- Computationally efficient.

Disadvantages

- Assumes feature independence.
- Performance may decrease when features are highly correlated.

2. Logistic Regression (9 Marks)

Logistic Regression is a supervised machine learning algorithm used for **classification problems**. It predicts the probability that an observation belongs to a particular class.

Used when the output is categorical.

Predicts probabilities between 0 and 1.

Commonly used for binary classification.

Uses the Sigmoid Function to convert outputs into probabilities.

Despite its name, it is a classification algorithm.

Sigmoid Function : $P(y) = \frac{1}{1+e^{-z}}$

Where:

- **P(y)** = Probability of belonging to a class
- **e** = Euler's constant
- **z** = Linear combination of input variables

Types of Logistic Regression

1. Binary Logistic Regression

Two classes.

Example: Fraud / Not Fraud

2. Multinomial Logistic Regression

More than two classes.

Example: Product categories

3. Ordinal Logistic Regression

Classes have a natural order.

Example: Low, Medium, High

Working of Logistic Regression (Short)

Step 1: Collect Training Data

Gather input features and corresponding class labels.

Step 2: Compute Linear Combination

Calculate weighted sum of input features.

Step 3: Apply Sigmoid Function

Convert the output into a probability value between 0 and 1.

Step 4: Calculate Probability

Estimate the probability of belonging to a class.

Step 5: Apply Threshold

Use a threshold (usually 0.5) to decide the class.

Step 6: Predict Class

Assign the final class label based on the probability.

Step 7: Evaluate Model

Measure performance using Accuracy, Precision, Recall, etc.

Applications

- Disease Prediction
- Customer Churn Prediction
- Fraud Detection
- Spam Detection
- Credit Risk Analysis

Advantages

- Easy to implement.
- Fast training and prediction.
- Produces probability scores.
- Easy to interpret.

Disadvantages

- Assumes linear relationship.
- Sensitive to outliers.
- Less effective for complex nonlinear data.

1. Clustering Methods (8 Marks)

Clustering is an **unsupervised machine learning technique** used to group similar data points into clusters. Objects within the same cluster are more similar to each other than to objects in other clusters. Does not require labeled data. Groups data based on similarity. Helps discover hidden patterns in data.

Widely used in Business Intelligence and data mining. Similar objects form a cluster, dissimilar objects belong to different clusters.

Working

1. Collect data.
2. Measure similarity between data points.
3. Form clusters based on similarity.
4. Refine clusters until optimal grouping is achieved.

Types of Clustering Methods

- Partition Clustering
- Hierarchical Clustering
- Density-Based Clustering
- Grid-Based Clustering

Applications

- Customer Segmentation
- Market Analysis
- Recommendation Systems
- Image Segmentation
- Document Clustering

Advantages

- Finds hidden patterns.
- No labeled data required.
- Useful for exploratory analysis.

Limitations

- Choosing the number of clusters can be difficult.
- Sensitive to noisy data.

Partition Methods (8 Marks)

Partition methods divide a dataset into a predefined number (**K**) of non-overlapping clusters, where each object belongs to exactly one cluster. Most common clustering approach. Requires the number of clusters in advance.

Minimizes distance within clusters. Maximizes distance between clusters.

Popular Algorithm: K-Means

1. Select K cluster centroids.
2. Assign data points to nearest centroid.
3. Recalculate centroids.
4. Repeat until clusters stabilize.

Characteristics

- Simple and efficient.
- Works well on large datasets.
- Produces compact clusters.

Advantages

- Easy to implement.
- Fast execution.
- Scalable to large datasets.

Limitations

- Need to specify K beforehand.
- Sensitive to outliers.
- May converge to local optimum.

3. Hierarchical Methods (8 Marks)

Hierarchical clustering creates a hierarchy of clusters represented as a tree-like structure called a **Dendrogram**.

Does not require specifying the number of clusters initially. Produces nested clusters.

Useful for understanding relationships among data points.

Types of Hierarchical Clustering

1. Agglomerative (Bottom-Up)

- Each data point starts as a separate cluster.
- Closest clusters are merged repeatedly.
- Continues until a single cluster remains.

2. Divisive (Top-Down)

- Starts with one large cluster.
- Repeatedly splits clusters into smaller groups.
- Continues until individual clusters are formed.

Working

1. Calculate similarity matrix.

2. Merge or split clusters.
3. Create dendrogram.
4. Select desired clustering level.

Advantages

- No need to specify K initially.
- Produces interpretable dendrogram.
- Useful for small datasets.

Limitations

- Computationally expensive.
- Difficult for very large datasets.

Evaluation of Clustering Models (8 Marks)

Evaluation of clustering models measures the quality and effectiveness of clusters formed by a clustering algorithm.

Need for Evaluation

- Determines cluster quality.
- Compares different clustering algorithms.
- Identifies the optimal number of clusters.
- Ensures meaningful grouping.

Evaluation Metrics

1. Silhouette Coefficient

Measures how similar an object is to its own cluster compared to other clusters.

- Value ranges from -1 to +1.
- Higher value indicates better clustering.

2. Davies-Bouldin Index (DBI)

Measures average similarity between clusters.

- Lower DBI indicates better clustering.

3. Calinski-Harabasz Index

Measures separation between clusters.

- Higher value indicates better clustering.

4. Within Cluster Sum of Squares (WCSS)

Measures compactness of clusters.

- Lower WCSS indicates better clusters.

Evaluation Criteria

- Cluster Cohesion (objects within cluster should be similar)

- Cluster Separation (different clusters should be distinct)
- Compactness
- Stability

Advantages

- Helps select the best clustering model.
- Measures clustering effectiveness.
- Improves decision-making.

Limitations

- Different metrics may give different results.
- No single metric works best for all datasets.

Association Rule (8 Marks)

Association Rule Mining is a data mining technique used to discover interesting relationships, patterns, or associations among items in large datasets. It is widely used in Business Intelligence for market basket analysis.

- Finds relationships between items that frequently occur together.
- Helps identify customer buying patterns.
- Discovers hidden patterns in transactional data.
- Generates rules in the form **If-Then** statements.

Structure of Association Rule

An association rule is represented as:

$$X \rightarrow Y$$

Where:

- **X (Antecedent)** = If part (condition)
- **Y (Consequent)** = Then part (result)

Example

$$\text{Bread} \rightarrow \text{Butter}$$

Meaning: Customers who buy bread are likely to buy butter.

Measures of Association Rules

1. Support

Measures how frequently an itemset appears in the dataset.

$$\text{Support}(X \rightarrow Y) = \frac{\text{Transactions containing } X \text{ and } Y}{\text{Total Transactions}}$$

2. Confidence

Measures how often Y occurs when X occurs.

$$\text{Confidence}(X \rightarrow Y) = \frac{\text{Support}(X \cup Y)}{\text{Support}(X)}$$

3. Lift

Measures the strength of association between X and Y.

$$\text{Lift} = \frac{\text{Confidence}(X \rightarrow Y)}{\text{Support}(Y)}$$

- Lift > 1 : Positive association
- Lift = 1 : Independent items
- Lift < 1 : Negative association

Working

1. Collect transaction data.
2. Find frequent itemsets.
3. Generate association rules.
4. Calculate support, confidence, and lift.
5. Select strong rules.

Applications

- Market Basket Analysis
- Product Recommendation
- Cross Selling
- Customer Behavior Analysis
- Inventory Planning

Advantages

- Discovers hidden patterns.
- Improves marketing strategies.
- Supports decision-making.

Limitations

- Generates a large number of rules.
- Can be computationally expensive.

Apriori Algorithm (8 Marks)

Apriori is a popular association rule mining algorithm used to find frequent itemsets and generate association

rules from transaction databases. Developed by Rakesh Agrawal and Ramakrishnan Srikant.

Uses the Apriori Principle: "If an itemset is frequent, all of its subsets must also be frequent."

Working of Apriori Algorithm

Step 1: Scan Database

Count the occurrence of each item.

Step 2: Generate Frequent 1-Itemsets

Select items whose support is greater than the minimum support threshold.

Step 3: Generate Candidate Itemsets

Combine frequent itemsets to create larger itemsets.

Step 4: Prune Infrequent Itemsets

Remove itemsets that do not satisfy minimum support.

Step 5: Repeat

Continue generating and pruning itemsets until no new frequent itemsets are found.

Step 6: Generate Association Rules

Calculate confidence and generate strong association rules.

Example

Transactions:

Transaction ID Items

T1	Bread, Milk
T2	Bread, Butter
T3	Milk, Butter
T4	Bread, Milk, Butter

Frequent itemsets are identified first, then rules such as:

Bread → Butter are generated.

Advantages

- Simple and easy to understand.
- Efficient pruning reduces search space.
- Widely used in data mining.

Limitations

- Requires multiple database scans.
- Computationally expensive for large datasets.
- Generates many candidate itemsets.